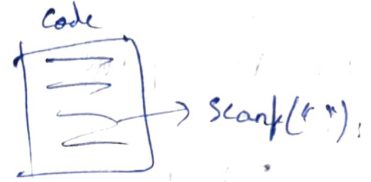
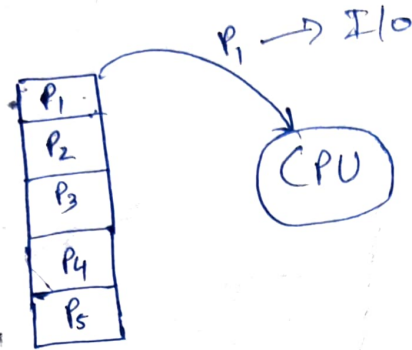


CPU Scheduling:-

①



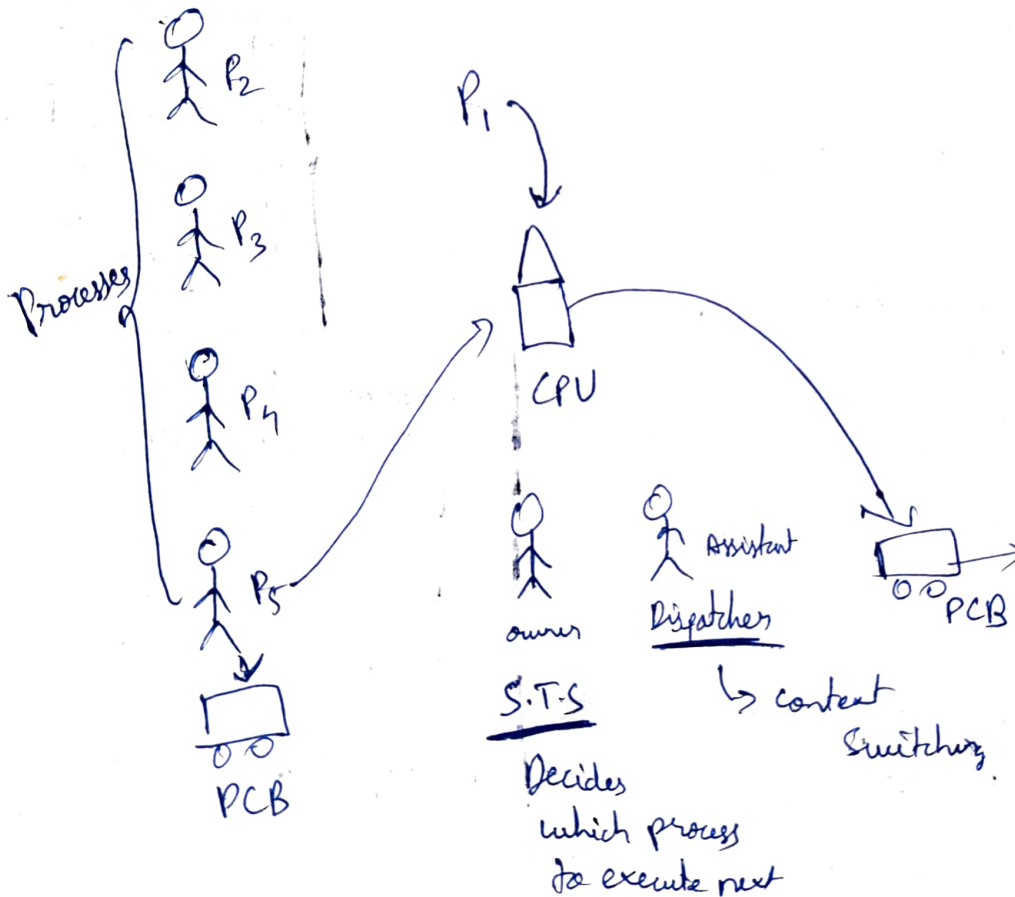
MIPS - Million instruction per second

Each instruction takes 1 tick

1 GHz $\rightarrow$ Clock frequency
5 GHz

$\rightarrow 10^9$ cycles/sec

$10^{-9} \rightarrow 1$ nanosecond
 $\rightarrow 1$ tick



CPU Scheduling Algorithm - STB

②

Non Preemptive

- FCFS
- SJF/SJN
- LSF/LJN
- MRRN

Preemptive

- Round Robin
- SRTF/DSRTN
- LRTF/LRTN
- Multilevel Queue Scheduling
- Multilevel Feedback Queue Scheduling

FCFS [First Come First Serve] :

Pid	BT (Burst time)	AT
P ₁	24	0
P ₂	3	0
P ₃	3	0

Tie breaker - Pid

→ Tie

Gantt Chart :-

P ₁	P ₂	P ₃
0	24	27 30

RT - Response time
WT - Waiting time
TAT - Turn Around Time
PFT - Process Finish Time
PAT - Process Arrival Time
BT - Burst Time
AT - Arrival Time

Turnaround Time :-

$$TAT = PFT - PAT$$

$$P_1 \quad 24 - 0 = 24$$

$$P_2 \quad 27 - 0 = 27$$

$$P_3 \quad 30 - 0 = 30$$

$$\text{Average TAT} = \frac{24 + 27 + 30}{3} = 27$$

Waiting Time :- $WT = TAT - BT$

$$P_1 \quad 24 - 24 = 0$$

$$P_2 \quad 27 - 3 = 24$$

$$P_3 \quad 30 - 3 = 27$$

$$\text{Average WT} = \frac{0 + 24 + 27}{3} = 17$$

Note :- Waiting time is the ~~process~~ time waited by the processes for the CPU.

Response Time: $RT = PST - PAT$ (Process start time - Process Arrival Time)

$P_1 \quad 0 - 0 = 0$

$P_2 \quad 24 - 0 = 24$

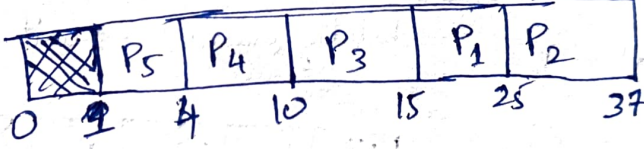
$P_3 \quad 27 - 0 = 27$

Average $RT = \frac{0 + 24 + 27}{3} = 17$

Note Waiting Time & Response Time are same & equal in case of Non preemptive scheduling

Throughput = $\frac{\text{No of processes}}{\text{Total completion time}} = \frac{3}{30} = \frac{1}{10} = 0.1$

CPU utilization = $\frac{\sum BT}{\text{Completion time of all processes}} \times 100 = \frac{30}{30} \times 100 = 100\%$

Pid	AT	BT	
P_1	5	10	
P_2	6	12	
P_3	3	5	
P_4	2	6	
P_5	1	3	

Waiting Time: $WT = TAT - BT$

P_1	$20 - 10 = 10$
P_2	$31 - 12 = 19$
P_3	$12 - 5 = 7$
P_4	$8 - 6 = 2$
P_5	$3 - 3 = 0$

Avg $WT = \frac{38}{5} = 7.6$

Response Time: $RT = PST - PAT$

P_1	$15 - 5 = 10$
P_2	$25 - 6 = 19$
P_3	$10 - 3 = 7$
P_4	$4 - 2 = 2$
P_5	$1 - 1 = 0$

Avg $RT = \frac{38}{5} = 7.6$

Turn Around Time

$TAT = PFT - PAT$

$P_1 \quad 25 - 5 = 20$

$P_2 \quad 37 - 6 = 31$

$P_3 \quad 15 - 3 = 12$

$P_4 \quad 10 - 2 = 8$

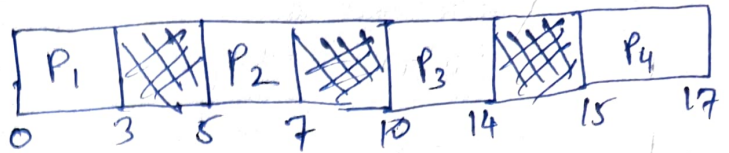
$P_5 \quad 4 - 1 = 3$

Average $TAT = \frac{74}{5} = 14.8$

CPU utilization = $\frac{\sum BT}{TCT} \times 100 = \frac{36}{37} \times 100 = 97.29\%$

Throughput = $\frac{\text{No of processes}}{TCT} = \frac{5}{37}$

Pid	AT	BT
P ₁	0	3
P ₂	5	2
P ₃	10	4
P ₄	15	2



(4)

Turn Around Time

$$TAT = PFT - PAT$$

$$P_1 \quad 3 - 0 = 3$$

$$P_2 \quad 7 - 5 = 2$$

$$P_3 \quad 14 - 10 = 4$$

$$P_4 \quad 17 - 15 = 2$$

$$\text{Avg TAT} = \frac{11}{4}$$

Waiting time

$$WT = TAT - BT$$

$$P_1 \quad 3 - 3 = 0$$

$$P_2 \quad 2 - 2 = 0$$

$$P_3 \quad 4 - 4 = 0$$

$$P_4 \quad 2 - 2 = 0$$

$$\text{Avg WT} = 0$$

Response Time

$$RT = PST - PAT$$

$$P_1 \quad 0 - 0 = 0$$

$$P_2 \quad 5 - 5 = 0$$

$$P_3 \quad 10 - 10 = 0$$

$$P_4 \quad 15 - 15 = 0$$

$$\text{Avg RT} = 0$$

$$\text{Throughput} = \frac{\text{No of processes}}{TCT}$$

$$= \frac{4}{17}$$

$$\text{CPU Utilization} = \frac{\sum BT}{TCT} \times 100$$

$$= \frac{11}{17} \times 100$$

Pid AT BT Scheduling overhead = 1 unit

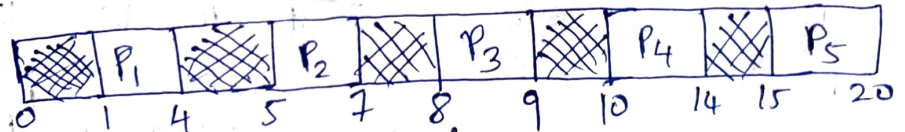
$$P_1 \quad 0 \quad 3$$

$$P_2 \quad 1 \quad 2$$

$$P_3 \quad 2 \quad 1$$

$$P_4 \quad 3 \quad 4$$

$$P_5 \quad 4 \quad 5$$



Turn Around Time

$$TAT = PFT - PAT$$

$$P_1 \quad 4 - 0 = 4$$

$$P_2 \quad 7 - 1 = 6$$

$$P_3 \quad 9 - 2 = 7$$

$$P_4 \quad 14 - 3 = 11$$

$$P_5 \quad 20 - 4 = 16$$

$$\text{Avg TAT} = \frac{44}{5}$$

Waiting Time

$$WT = TAT - BT$$

$$P_1 \quad 4 - 3 = 1$$

$$P_2 \quad 6 - 2 = 4$$

$$P_3 \quad 7 - 1 = 6$$

$$P_4 \quad 11 - 4 = 7$$

$$P_5 \quad 16 - 5 = 11$$

$$\text{Avg WT} = \frac{29}{5}$$

Response Time

$$RT = PST - PAT$$

$$P_1 \quad 1 - 0 = 1$$

$$P_2 \quad 5 - 1 = 4$$

$$P_3 \quad 8 - 2 = 6$$

$$P_4 \quad 10 - 3 = 7$$

$$P_5 \quad 15 - 4 = 11$$

$$\text{Avg RT} = \frac{29}{5}$$

$$\text{Throughput} = \frac{\text{No of processes}}{TCT}$$

$$= \frac{5}{20}$$

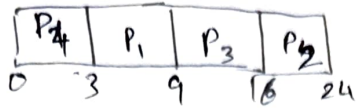
$$\text{CPU Utilization} = \frac{\sum BT}{TCT} \times 100$$

$$= \frac{15}{20} \times 100$$

Shortest Job First (SJF) / Short Job Next (SJN):

(5)

Pid	BT
P ₁	6
P ₂	8
P ₃	7
P ₄	3



$$TAT = PFT - PAT$$

P ₁	9
P ₂	24
P ₃	16
P ₄	3

$$WT = TAT - BT$$

P ₁	3
P ₂	16
P ₃	9
P ₄	0

$$Avg TAT = 13$$

$$Avg WT = 7$$

$$RT = PST - PAT$$

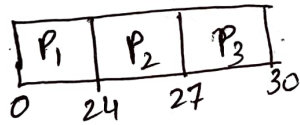
P ₁	3
P ₂	16
P ₃	9
P ₄	0

$$Throughput = \frac{\text{No. of process}}{T.CT} = \frac{4}{24} = \frac{1}{6}$$

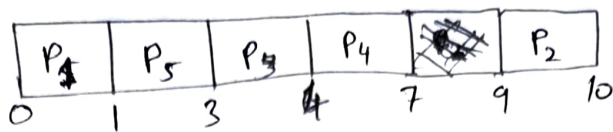
$$CPU \text{ utilization} = \frac{\sum BT}{TCT} \times 100 = \frac{24}{24} = 100\%$$

$$Avg RT = 7$$

Pid	AT	BT
P ₁	0	24
P ₂	1	3
P ₃	2	3

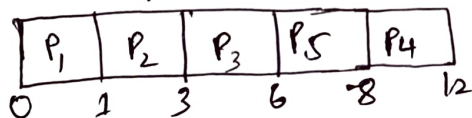


Pid	AT	BT
P ₁	0	1
P ₂	9	1
P ₃	3	1
P ₄	2	3
P ₅	1	2



6

Pid	AT	BT
P ₁	0	1
P ₂	1	2
P ₃	2	3
P ₄	3	4
P ₅	4	2



7

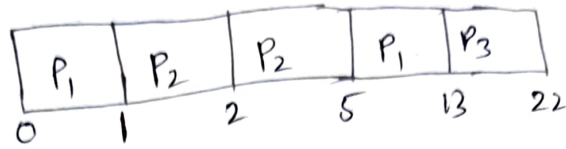
Diagram illustrating a sequence of operations on a data structure (likely a linked list or array) with indices 0 to 26. The sequence is: P_1 , P_2 , P_2 , P_2 , P_4 , P_1 , P_3 .

P_1	P_2	P_3	P_4	P_5	P_4	P_3	P_2	P_1
0	2	3	8	10	12	15	25	67

gate 2011

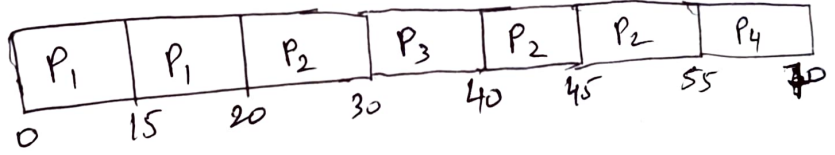
8

Pid	AT	BT
P ₁	0	9 8
P ₂	1	4 3
P ₃	2	9



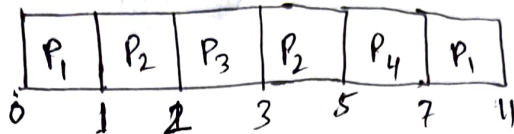
gate 2007

Pid	AT	BT
P ₁	0	20 5
P ₂	15	25 15 10
P ₃	30	10
P ₄	45	15



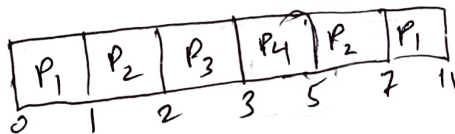
SRTF

Pid	AT	BT
P ₁	0	5 4
P ₂	1	3 2
P ₃	2	1 0
P ₄	3	2



(9)

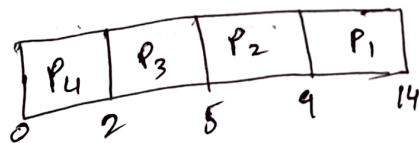
Pid	AT	BT
P ₁	0	5 4
P ₂	1	3 2
P ₃	2	1
P ₄	3	2



Tie Breaker:-

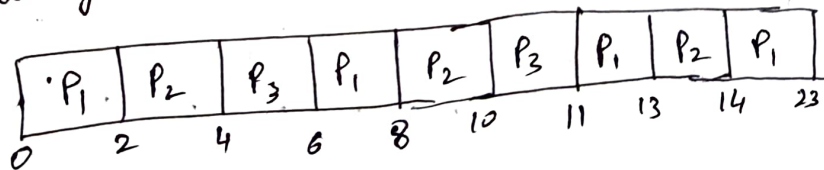
Priority is given to process that is not started

Pid	BT
P ₁	5
P ₂	4
P ₃	3
P ₄	2



Round Robin Scheduling :- FCFS preemptive

Pid	BT	TQ=2
P ₁	15	13 11 9
P ₂	5	3 1
P ₃	3	1



AT BT TQ=2

$\frac{P_1}{0}$
 $\frac{P_2 P_1}{2}$
 $\frac{P_1 P_3 P_2}{4}$
 $\frac{P_3 P_2 P_1}{6}$

P ₁	P ₂	P ₁	P ₃	P ₂	P ₁
0	2	4	6	8	10

P ₁	P ₂	P ₁	P ₃	P ₂	P ₄	P ₁	P ₃
0	2	4	6	8	10	12	14

Pid AT BT TQ=3

$\frac{P_1}{0}$
 $\frac{P_2 P_1}{3}$
 $\frac{P_1 P_3 P_4 P_2}{6}$

P ₁	P ₂	P ₁	P ₃	P ₄	P ₂	P ₁	P ₃	P ₄
0	3	6	8	10	12	14	16	18

P ₂	P ₁	P ₃	P ₄	P ₂
20	22	23	25	27

Pid AT BT TQ=3

$\frac{P_1}{0}$
 $\frac{P_2 P_1}{3}$
 $\frac{P_1 P_3 P_4 P_2}{6}$

P ₁	P ₂	P ₁	P ₃	P ₄	P ₂	P ₁	P ₃	P ₄	P ₂
0	3	6	9	12	15	18	20	23	26

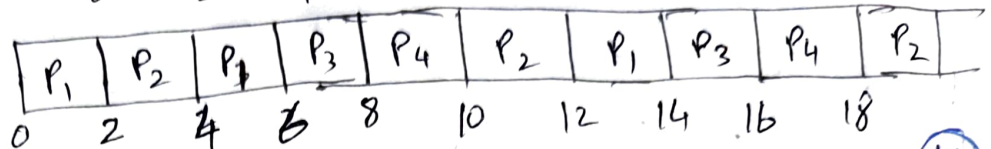
P ₃	P ₄	P ₃	P ₄
29	32	35	38

Pid	AT	BT
P ₁	0	8 6 4 2 0
P ₂	1	6 4 2 0
P ₃	3	5 3 1
P ₄	4	4 2 0

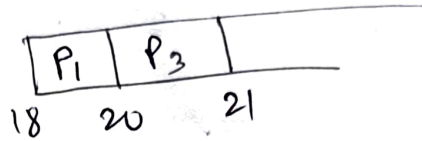
TQ=2

$$\frac{P_1}{0} \quad \frac{P_2 P_1}{2} \quad \frac{P_1 P_2 P_4 P_2}{4}$$

The breaker - process which is not executed & given priority.

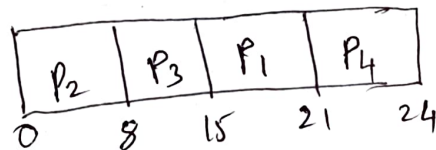


(11)

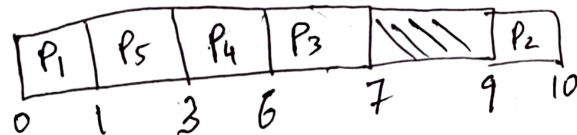


Longest Job First (LJF)

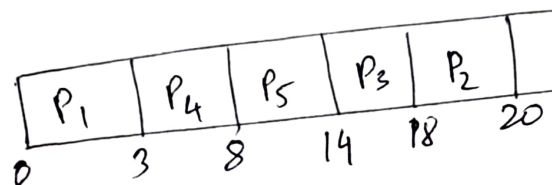
Pid	BT
P ₁	6
P ₂	8
P ₃	7
P ₄	3



Pid	AT	BT
P ₁	0	1
P ₂	9	1
P ₃	3	1
P ₄	2	3
P ₅	1	2



Pid	AT	BT
P ₁	0	3
P ₂	1	2
P ₃	2	4
P ₄	3	5
P ₅	4	6

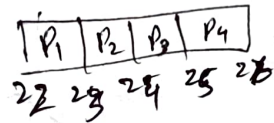
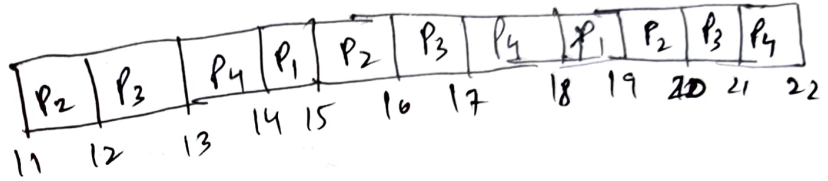
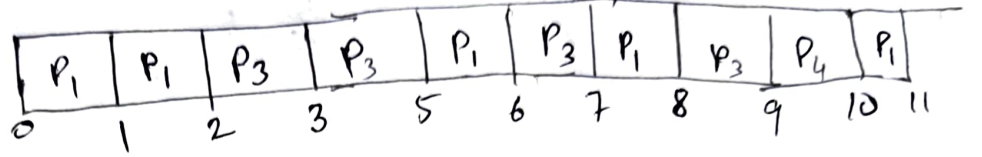


LRTF/LRTN

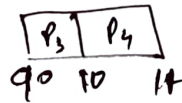
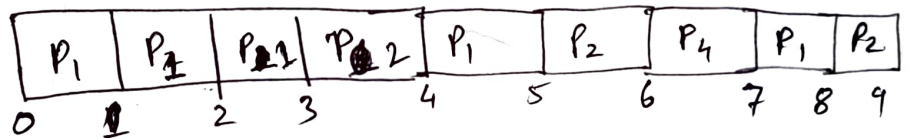
Tie breaker - AT

(12)

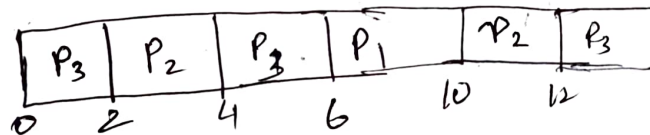
Pid	AT	BT
P ₁	0	8 7 6 5 4
P ₂	1	4
P ₃	2	9 8 6 5 4
P ₄	3	5 4



Pid	AT	BT
P ₁	0	5 4 3 2 1
P ₂	1	3 2 1
P ₃	2	1 0
P ₄	3	2



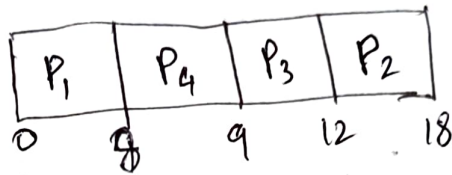
Pid	BT	AT
P ₁	2 0	0
P ₂	4 2	0
P ₃	6 4 2	0



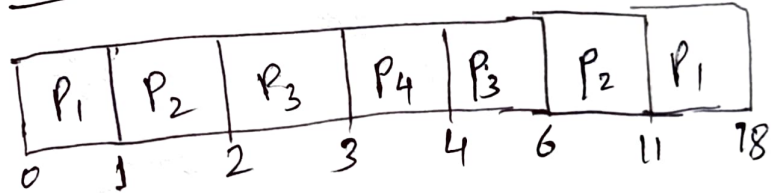
Priority Scheduling - $\rightarrow$ NP

Pid	AT	BT	Priority
P ₁	0	87	4
P ₂	1	65	3
P ₃	2	32	2
P ₄	3	10	1

Non Preemptive:

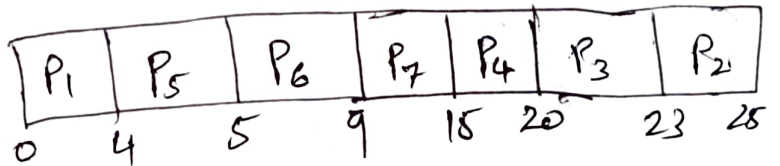


Preemptive:

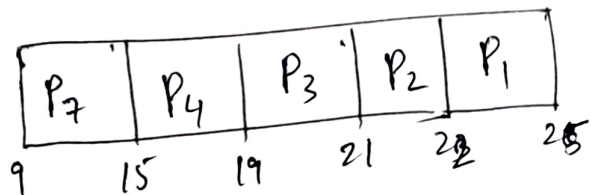
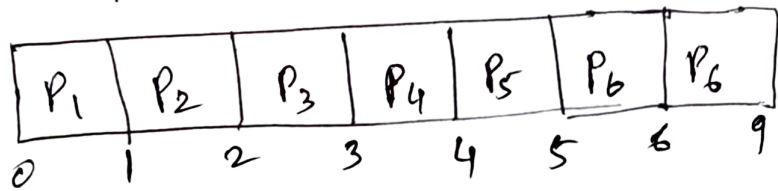


Pid	Priority	AT	BT
P ₁	2 (L)	0	43
P ₂	4	1	21
P ₃	6	2	32
P ₄	8	3	54
P ₅	10	4	10
P ₆	12 (H)	5	43
P ₇	9	6	6

Non Preemptive:



Preemptive



Non - Preemptive only

HRRN :- Highest Response Ratio Next

(14)

Pid AT BT

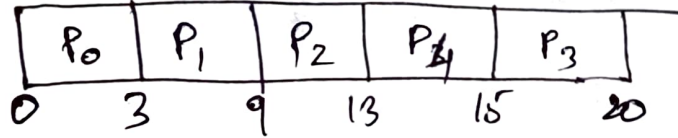
P₀ 0 3

P₁ 2 6

P₂ 4 4

P₃ 6 5

P₄ 8 2



$$\frac{WT + BT}{BT}$$

$$RR_2 = \frac{5 + 4}{4} = 2.25$$

$$RR_3 = \frac{3 + 5}{5} = 1.4$$

$$RR_4 = \frac{1 + 2}{2} = 1.5$$

$$RR_3 = \frac{7 + 5}{5} = 2.4$$

$$RR_4 = \frac{8 + 2}{2} = 3.5$$